## General commands:

### #Help pages:

```
$ help
$ help [command]
```

### #Save files:

$ save scaffold.pdb *model-spec*
$ save hybrid_1.mrc *model-spec*
$ save chimerax.cxs
$ save coordinates.pdb *model-spec* displayedOnly true selectedOnly True

### #Fetch files from various online databases:

```
$ open 4893 fromDatabase emdb
$ open 6ALH fromDatabase pdb
```

## 1 Atom and object selection syntax:

### #Select residues in model 1, chains I and J:

$ select #1/i:830-945,1037-1058/j:24-112
$ select (add | subtract | intersect) *spec*
$ select @H     --select all hydrogens

### #Save selections:

```
$ name frozen mcore sel
$ name delete mcore               --forgets the selection
```

### #Deselect all:

```
$ ~select
```

### #Select residues of all currently selected atoms:

```
$ select sel residues true
```

### #Show only bonds from salt bridges:

```
$ hbonds sel reveal true showDist true saltOnly true
```

## 3 Volume operations:

### #Extract map density around atoms:

```
$ color zone #5 near #4 distance 5
$ volume splitbyzone #5
```

**#Volume resampling:**

```
$ volume resample #1 onGrid #4
$ volume zone #6 nearAtoms #5 range 5
```

**#Change volume step and level (all maps):**

```
$ volume step 1
$ volume level 0.1
```

**#Hide disconnected density:**

```
$ surface dust #2 size 5
```

**#Scale maps and calculate difference:**

```
$ volume subtract #1 #3 minRMS True onGrid #1
```

**#Transparency: make (a)toms and (c)artoons 80% transparent:**

```
$ transparency #1 80 target ac
```

**#Depth cue:**

```
$ lighting depthCue true depthCueStart 0.4 depthCueEnd 0.7
```

**#Superpose two structures globally:**

```
$ mm #2 to #1 showAlignment True
```

**#Morph between two structures:**

```
$ morph #1,2 wrap true frames 60
```

**#Draw an icosahedron:**

```
$ shape icos divisions 1 mesh True radius 400
```

**#Hide cartoon rendering of chains g-k in model #1:**

```
$ hide #1/g-k cartoons
```

**#Align by LSQ:**

```
$ align #2 toAtoms #1 matchChainIds True reportMatrix True
```

**#Assign secondary structure (0=coil, 1=alpha helix, 2=beta strand):**

```
$ setattr #3/j:365-369 res ss_type 2
```

- **Movies:**

**#Open multiple maps as a volume series and play the series:**

```
$ open *.mrc vseries true
```

**#The colorRange option is for coloring the map with 'color zone':**

```
$ vseries play #1 direction oscillate normalize true loop true
  cacheFrames 60 colorRange 20
```

**#Rock object back and forth:**

```
$ turn y 25 300 rock 150
```

**#Record movie with 4 commands on one line separated by semicolons:**

```
$ movie record; turn y 25 300 rock 150; wait 300; movie encode
  ~/Desktop/movie2.mp4
```

**#Record a movie at high resolution with no motion:**

```
$ movie record size 4096,4096; wait 5; movie encode
  ~/Desktop/gbp.mp4 framerate 1
```

**#Rotate a model clockwise about a defined point along the x-axis:**

```
$ turn x -7 center 178.38,157.92,156.14 coordinateSystem 7
  models 9
```

**#Apply model position to another model:**

```
$ view position model-spec sameAsModels ref-model-spec
```

**#Print matrix of current camera and model position:**

```
$ view matrix
```

**#Apply transformation matrix:**

```
$ view matrix mod 1,1,0,0,0,0,1,0,0,0,0,1,0
```

**#CAV1 8S movie:**

```
$ movie record; turn x 1 90; wait 180; turn y 1 60; wait 180;
  turn x 1 90; wait 180; transparency #2 80; wait 180; turn x 1
  180; wait 180; movie encode quality highest output
  Cav1_8S-movie.mp4; movie stop
```

## 5. Coloring operations:

**#Apply colors from model to density map:**

```
$  color radial #1 center 0,0,0
$  color sel yellow
```

**#Sequentially color the atoms and cartoon with the following palette:**

```
$  color sequential #1 chains target ac palette
   6633FF:660099:CC66FF:9966FF:3300FF:6666FF:6699FF:0066FF:00
   CCFF:99CCFF:333399cre
```

**#CAV1 8S figure coloring scheme:**

```
$  color #1/a-k:49-60 #ffff33; color #1/a-k:61-67 #3399ff;
   color #1/a-k:68-75 #ff3333; color #1/a-k:76-80 #3399ff;
   color #1/a-k:81-101 #00e673; color #1/a-k:102-134 #9933ff;
   color #1/a-k:135-169 #999999; color #1/a-k:170-end #33ffff
```

## 6. Command files:

**#Open and run a ChimeraX command file (*.cxc):**

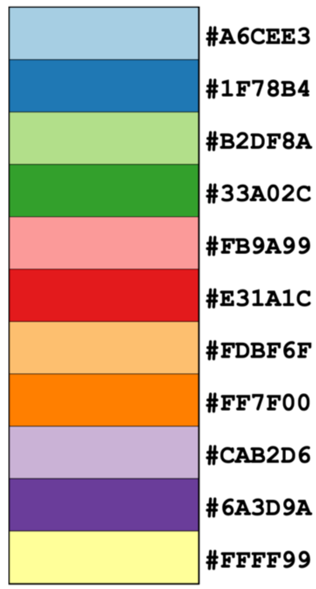
```
$  open initialize_commands.cxc
```

**#Example of a ChimeraX command file:**

```
#Initialize model and maps
#BEGIN
select #3; volume sel style surface
~select
hide #1 target s
wait 30
color #1/b-k #3399FF; color #1/a #FF3399
wait 30
show #!1 target ac; color #1 byhetero;
wait 30
hide #1 models
wait 30
show #2; show #3
wait 30
transparency #2 0; transparency #3 50
wait 30
view orient
#END
```

**5. ChimeraX color palettes from [colorbrewer.org:](colorbrewer.org)**

- 11-class paired: (colorbrewer 11-class paired):




- 11-class RdYIBu (colorbrewer 11-class RdYlIBu):
  - color blind friendly